

# Assessment Brief

## Submission details

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| <b>Module title:</b>                                   | Data Science for Cyber Security                                                                                                                                                                                                                                                                                                                                       |
| <b>Module code:</b>                                    | UFCE8B-30-3                                                                                                                                                                                                                                                                                                                                                           |
| <b>Module Leader:</b>                                  | Ahsan Ikram                                                                                                                                                                                                                                                                                                                                                           |
| <b>Assessment title:</b>                               | Data Science Portfolio                                                                                                                                                                                                                                                                                                                                                |
| <b>Year:</b>                                           | 2025-26                                                                                                                                                                                                                                                                                                                                                               |
| <b>Assessment type:</b>                                | Portfolio                                                                                                                                                                                                                                                                                                                                                             |
| <b>Assessment weighting:</b>                           | 100% of total module mark                                                                                                                                                                                                                                                                                                                                             |
| <b>Size or length of assessment:</b>                   | 3000-words or equivalent                                                                                                                                                                                                                                                                                                                                              |
| <b>Module learning outcomes assessed by this task:</b> | <ol style="list-style-type: none"> <li>1. To demonstrate understanding of programming for the purposes of data science in cyber security</li> <li>2. To demonstrate practical skills of gathering, processing and analysing data to solve cyber security challenges</li> <li>3. To reflect upon techniques and methods used and evaluate their performance</li> </ol> |

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| <b>Use of AI in the assessment:</b> | <p><b>Generative Artificial intelligence (Gen AI) tools can be used in an assistive role</b></p> <p>You are permitted to use Gen AI tools in an assistive role in this assessment to support you with</p> <ul style="list-style-type: none"> <li>• drafting and structuring content (<u>excluding software or coding content</u>)</li> <li>• supporting a particular process such as translating content</li> <li>• providing ideas or inspiration to help you overcome a creative block.</li> </ul> <p>Please note that while GenAI tools might suggest topics/content, you will always need to go back and locate and critically engage with actual literature/research/evidence yourself.</p> |
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As the use of Gen AI is optional (due to its purely supportive capacity) it's use is not part of the skills or knowledge you need to demonstrate for this assessment. However, if you have used Gen AI to prepare or produce part or all your work, you must acknowledge its use. For more advice on this (and its use in general) please see your [Generative AI study skills guide](#).

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guidance on Referencing (inc AI):</b> | <p>Please note that the aim of referencing is to demonstrate you have read and understood a range of sources to evidence your key points. You need to list the references consistently and in such a way as to ensure the reader can follow up on the sources for themselves.</p> <p><a href="#">Referencing - Study skills   UWE Bristol</a></p> <p><a href="#">Using generative AI at UWE Bristol - Study skills   UWE Bristol</a></p> |
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## Submission and feedback dates

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|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Submission deadline:</b>                     | Before 14:00 on 07/05/2026<br>Is eligible for 48-hour late submission window.<br><i>All times are 24-hour clock, current local time (at time of submission) in the UK.</i>                                                                                                                                                                                                                                                                                                              |
| <b>Submission format:</b>                       | You are required to submit a zipped (compressed) file via Blackboard. The file should include, <ol style="list-style-type: none"><li>1. A completed coversheet. (The template for the coversheet will be provided)</li><li>2. A sub-folder for each portfolio task. (Each folder should contain all relevant data and materials needed to run/validate the completed tasks).</li><li>3. A video file (or URL) demonstrating the working and execution of the completed tasks.</li></ol> |
| <b>Marks and Feedback due on:</b>               | 20 working days after the submission window date.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Marks and Feedback will be provided via:</b> | Blackboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Completing your assessment

### Where am I required to do on this assessment?

This is an individual assignment.

The assignment requires you to complete a set of tasks (04), combined into a portfolio. The tasks will be aimed at applying the data science principles and techniques covered in the module to cyber security related case-studies and datasets.

The tasks are briefly described in the table below. More specific details and datasets for the tasks will be released in the weeks indicated in the 'Week released' column.

| Task | Description                                                                                                                                                                                                                                                                               | Week released                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 01   | <b>Data Quality Pipeline Task:</b> Using the given dataset, create a data quality pipeline; the pipeline must include, data acquisition, data validation, data cleaning and at least two analysis and visualisations. Optionally you can consider transformations to add advanced phases. | To be released in <b>Week 4.</b>  |
| 02   | <b>Machine Learning Task:</b> Comparatively analyse and select a machine learning model to solve a cyber security case study, for example, phishing, bullying, intrusion etc.                                                                                                             | To be released in <b>Week 8.</b>  |
| 03   | <b>Threat/Anomaly Analysis Task:</b> Create an experimental proof of concept for applying data analytics to malware analysis.                                                                                                                                                             | To be released in <b>Week 12.</b> |
| 04   | <b>Security Monitoring Dashboard:</b> Select and apply data science metrics to a cyber security case study and prepare a monitoring dashboard to show the real-time analysis.                                                                                                             | To be released in <b>Week 16.</b> |

Working on this assignment will help you to practice and exercise applied data science skills in a cyber security context.

If you have questions about this assignment, please post them to the discussion board 'Assessment Queries & Answers' on Blackboard.

### Where should I start?

Focus on developing a clear understanding of key data science concepts and explore the cyber security domains where the data science concepts can be applied.

### What do I need to do to pass?

You need to show evidence that you have attempted and completed at least half of the portfolio tasks to a good standard to pass the module.

### How do I achieve high marks in this assessment?

Complete all the tasks included in the portfolio. Provide clear, well-structured and thoroughly documented and tested codebase. Prepare the final portfolio report to a high standard as highlighted in the marking criteria.

### How does the learning and teaching relate to the assessment?

In the first half of the module, we extensively worked on learning and applying core data science concepts. In the second half we aimed to apply the data science concepts to cyber security problems/datasets.

### What additional resources may help me complete this assessment?

FAQs and other resources on your Blackboard site

Specific UWE library study skills pages i.e.

- <https://www.uwe.ac.uk/study/study-support/study-skills>
- [Critical thinking and writing - Reading and writing | UWE Bristol](#)
- [Writing - Reading and writing | UWE Bristol](#) consider linking to specific assessment types from this link
- [How to give a presentation](#)
- [Writing feedback - Writing | UWE Bristol](#)
- [Studiosity - Study support | UWE Bristol](#) for quick feedback on your writing

### What do I do if I am concerned about completing this assessment?

It is recommended that you review all of the relevant materials on Blackboard. You can also speak to your module leader for advice and guidance.

UWE Bristol offer a range of Assessment Support Options that you can explore through [this link](#), and both [Student Support Advisers](#) and [Wellbeing Support](#) are available.

For further information, please see the [Student study essentials](#).

#### **How do I avoid an Assessment Offence on this module?**

Use the support above if you feel unable to submit your own work for this module.

Outline the most common forms of AOs for this type of assessment and brief guidance on how to avoid this.

Refer to [academic-conduct-policy-and-academic-misconduct-procedures](#)

## Marks and Feedback

Your assessment will be marked according to the following marking criteria.

You can use these to evaluate your own work before you submit.

The following marking criteria will be applied to the entire assessment. The 'Accuracy & Completeness' criteria will be further defined with the release of each portfolio task.

| Percentage mark                                                                                                                                         | 86-100                                                                                                                                                                                                                                     | 70-85                                                                                                                                                                                                                                  | 60-69                                                                                                                                                                                                                   | 50-59                                                                                                                                                                                                                                         | 40-49                                                                                                                                                                                                                       | 0-40                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Overall Descriptor                                                                                                                                      | Outstanding                                                                                                                                                                                                                                | Excellent                                                                                                                                                                                                                              | Very good                                                                                                                                                                                                               | Good                                                                                                                                                                                                                                          | Adequate                                                                                                                                                                                                                    | Inadequate                                                                                                      |
| Assessment criteria                                                                                                                                     | PASS                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                        |                                                                                                                                                                                                                         |                                                                                                                                                                                                                                               |                                                                                                                                                                                                                             |                                                                                                                 |
| <b>Accuracy &amp; Completeness (80%)</b><br><br><i>20% per portfolio task. Further marking criteria discussed within portfolio task specifications.</i> | All portfolio tasks completed to a high standard, results are verified and validated. Codebase is structured and executable.<br><br>Performance and accuracy metrics considered throughout.                                                | All portfolio tasks attempted and mostly completed; results are mostly verified and validated. Codebase is structured and executable.<br><br>Performance and accuracy metrics considered partially.                                    | Most of the portfolio tasks attempted and completed, results are occasionally verified and validated. Codebase is partially structured and executable.<br><br>Performance and accuracy metrics are sparsely considered. | At least half of the portfolio tasks attempted and completed, results are rarely verified and validated. Codebase is rarely structured and executable.<br><br>Performance and accuracy metrics are not considered.                            | Less than half of the portfolio tasks attempted and completed, results are not verified and validated. Codebase is not structured and executable.<br><br>Performance and accuracy metrics are not considered.               | Portfolio tasks barely attempted; results unclear, codebase not working.                                        |
| <b>Video demonstration (15%)</b>                                                                                                                        | Comprehensive video demonstration with clear and informative commentary. The demonstration covers all tasks in context of working and structure. The demonstration shows execution of the tasks covering all possible scenarios/use cases. | Thorough video demonstration with clear and informative commentary. The demonstration covers all tasks in context of working and structure. The demonstration shows execution of the tasks covering most/required scenarios/use cases. | Good video demonstration with relevant commentary. The demonstration covers most tasks in context of working and structure. The demonstration shows execution of the tasks in at least one scenario/use case.           | Sufficiently well-presented video demonstration with some commentary. The demonstration covers at least 50% tasks in context of working and structure. The demonstration shows execution some of the tasks in at least one scenario/use case. | Video demonstration lacks quality and completeness. The demonstration covers at least 50% tasks in context of working and structure. The demonstration shows execution some of the tasks in at least one scenario/use case. | Incomplete or missing video demonstration.                                                                      |
| <b>Presentation, citations &amp; references (5%)</b>                                                                                                    | Exceptional style, standard of writing & presentation; may be publishable. All sources correctly cited with full details listed                                                                                                            | Very well-written & presented, negligible errors in typography or style. All sources correctly cited & listed; errors in style or detail only                                                                                          | Very well-written & presented with only minor errors in typography or style. Sources largely cited & listed correctly; errors in style or detail only                                                                   | Reasonably well-written & presented; scope to improve style or proof-reading. Most sources cited & listed; some details missing/ incomplete                                                                                                   | Layout very basic; readability marred by errors in use of English/ proof-reading. Some citation of sources but some source details missing/ incomplete                                                                      | Structure poor; no real argument discernible. Layout poor; grammatical, spelling/ punctuation errors throughout |

1. In line with UWE Bristol's [Assessment Content Limit Policy](#) (formerly the Word Count Policy), word count includes all text, including (but not limited to): the main body of text (including headings), all citations (both in and out of brackets), text boxes, tables and graphs, figures and diagrams, quotes, lists.
2. UWE Bristol's Academic Conduct Policy and Academic Misconduct Procedures (Assessment Offence Policy) is available from the [Academic Integrity](#) webpages and requires that you submit work that is entirely your own and reflects your own learning, so it is important to:
  - Ensure you reference all sources used, using the [UWE Harvard/OSCOLA](#) system (amend as appropriate) and the guidance available on [UWE's Study Skills referencing pages](#).
  - Refer to peer reviewed primary sources, rather than using AI or secondary sources
  - Avoid copying and pasting any work into this assessment, including your own previous assessments, work from other students or internet sources
  - Develop your own style, arguments and wording, so avoid copying sources and changing individual words but keeping, essentially, the same sentences and/or structures from other sources
  - Never give your work to others who may copy it
  - If an individual assessment, develop your own work and preparation, and do not allow anyone to make amends on your work (including proof-readers, who may highlight issues but not edit the work).

**When submitting your work, you will be required to confirm that the work is your own**, and text-matching software and other methods are routinely used to check submissions against other submissions to the university and internet sources. Details of what constitutes plagiarism and how to avoid it can be found on UWE's Study Skills [pages about avoiding plagiarism](#).

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